FOR A PROJECTIVE MODULE THE HOMOTHETIES AND THE LOCALIZATION MAP OF THE HOMOTHETY SITUATION ARE UNIVERSALLY INJECTIVE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The spectrum of the homotheties and the notation Σ, N used in the statement. Same chapter flat.tex, label lemma-homothety-spectrum (statement line 1978). It fixes the multiplicative subset Σ and the module N on which the projectivity hypothesis is imposed.
- Universal injectivity of a homothety in the local case. Same chapter flat.tex, label lemma-universally-injective-local (invoked at proof line 1987). Applied to each $s \in \Sigma$, it gives universal injectivity of $s : N \to N$ over R.

Source reference. Original lemma 05DE: sources/stacks-flat/pinned-original-source.tex, source lines 1783–1783. Navigation pointer for \ref{lemma-homothety-spectrum} used inside the counted span (line 1978); the argument that consumes it is retained in the span.

Source reference. Original lemma 05FQ: sources/stacks-flat/pinned-original-source.tex, source lines 1946–1946. Navigation pointer for \ref{lemma-universally-injective-local} used inside the counted span (line 1987); the argument that consumes it is retained in the span.

Lemma 0.1. *Assumption and notation as in Lemma 05DE. Assume moreover that N is projective as an R-module. Then each $s \in \Sigma$ defines a universally injective R-module map $s : N \to N$, and the map $N \to \Sigma^{-1}N$ is R-universally injective.*

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Proof. Pick $s \in \Sigma$. By Lemma 05FQ the map $s : N \to N$ is universally injective. The map $N \to \Sigma^{-1}N$ is universally injective as the directed colimit of the maps $s : N \to N$. $\square$